

Requirements Analysis — GameStore Platform

1. Domain Model — Entity Catalog

The GameStore platform comprises 23 entities mapped from the real codebase (GameStore.DAL.Entities).

Entity List

USER	GAME	CART	CARTITEM	LIBRARY	LIBRARYGAME	ORDER	ORDERITEM	REVIEW
WISHLISTITEM	FRIENDSHIP	MESSAGE	CATEGORY	SUPPORTTICKET	TICKETREPLY	PAYOUT		
COUPON	NOTIFICATION	AUDITLOG	REFRESHTOKEN	BANLOG	REPORT	DEVELOPERPROFILE		

Enum Catalog

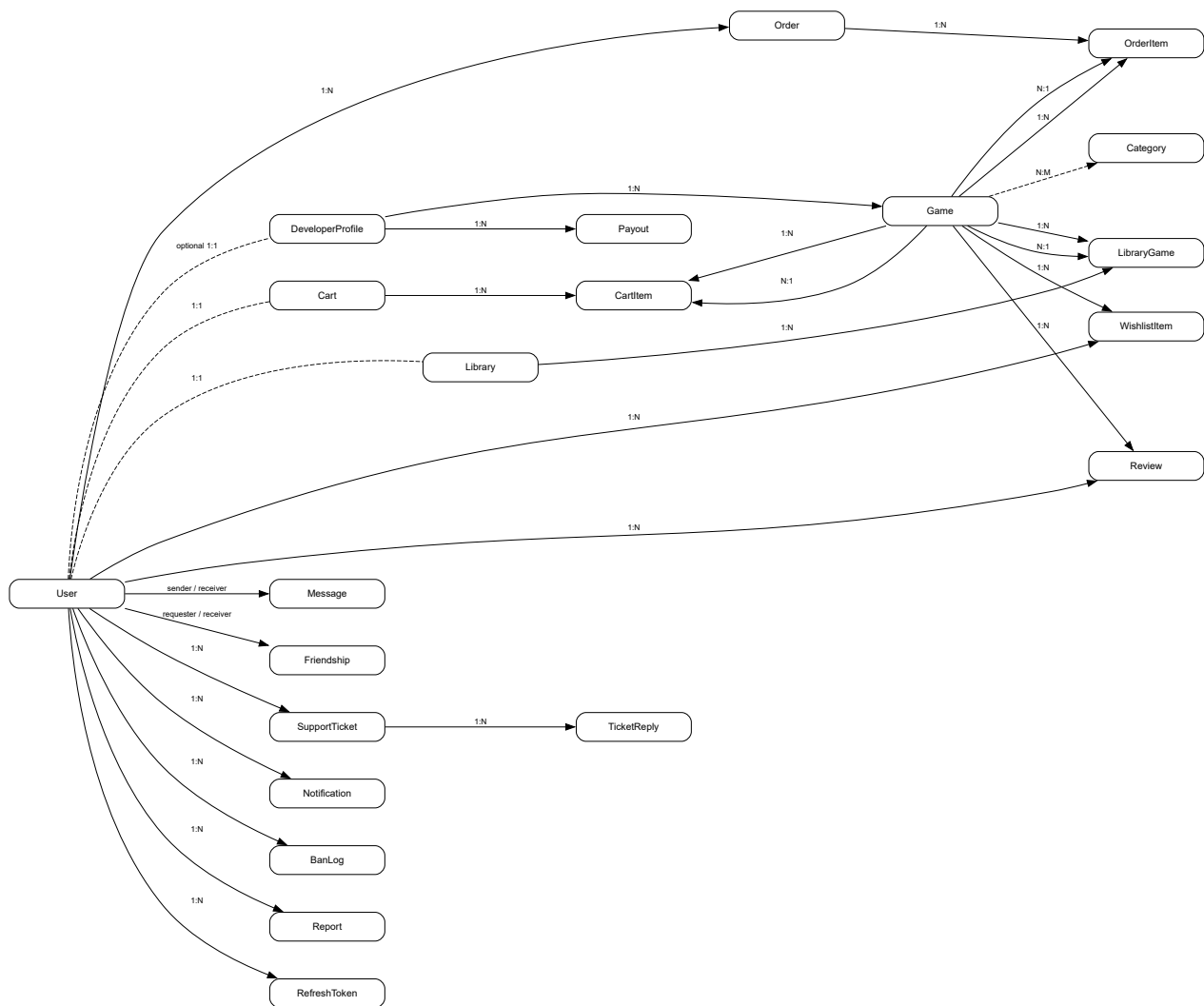
Enum	Values
Role	CUSTOMER, DEVELOPER, ADMIN
GameStatus	DRAFT, PENDING, APPROVED, REJECTED, ARCHIVED
OrderStatus	PENDING, COMPLETED, CANCELLED, REFUNDED
TicketStatus	OPEN, IN_PROGRESS, RESOLVED, CLOSED

Key Entity Relationships

Entity	Relationships
User	Has one Cart (1:1), one Library (1:1), many WishlistItems (1:N), many Reviews (1:N), sent/received Messages, sent/received FriendRequests, many Orders, many Notifications,

Entity	Relationships
	optional DeveloperProfile (1:1), many Reports, many BanLogs, many Payouts, many SupportTickets
Game	Belongs to a Developer (User via DeveloperProfile), has many Categories (N:M via GameCategory), many CartItems, many LibraryGames, many OrderItems, many Reviews, many WishlistItems
Cart	Belongs to one User (1:1), contains many CartItems (1:N) → each CartItem points to one Game
Library	Belongs to one User (1:1), contains many LibraryGames (1:N) → each LibraryGame points to one Game
Order	Belongs to one User (1:N), contains many OrderItems (1:N) → each OrderItem points to one Game with price-at-purchase and commission
SupportTicket	Belongs to one User (optional, 1:N), contains many TicketReplies (1:N)
Friendship	Links two Users (Requester + Receiver) with status (Pending / Accepted / Rejected)
Message	Sent by one User to another User with read tracking
AuditLog	Tracks entity changes across the system for compliance

Entity-Relationship Diagram



2. Stakeholder Personas

Customer (Omar)

Role: End-user who browses, purchases, and plays games.

Goals: Discover new games, buy securely, build library, connect with friends.

Pain Points: Unclear refund policy, difficulty finding specific games, payment issues.

Developer (Sara)

Role: Game creator who uploads and sells games on the platform.

Goals: Publish games easily, track sales analytics, receive timely payouts.

Pain Points: Complex upload process, delayed payout cycles, limited visibility into sales data.

Admin (Ahmed)

Role: Platform manager responsible for moderation, user management, and system health.

Goals: Efficient content moderation, accurate revenue reporting, fraud detection.

Pain Points: Fake or inappropriate game submissions, fraudulent transactions, manual approval bottlenecks.

3. Role & Permission Matrix

CUSTOMER

- Browse and search game catalog
- View game details and reviews
- Purchase games via Stripe or store credit
- Write, edit, and delete reviews
- Add/remove games from wishlist
- View owned games in library
- Download purchased games
- Send/receive friend requests
- Chat with friends (SignalR)
- View and manage notifications
- Submit and track support tickets
- Apply coupons at checkout
- Request refunds
- Edit profile, change password
- Post on profiles

DEVELOPER

- All CUSTOMER permissions
- Create developer profile (studio)
- Upload and manage game listings
- Edit game metadata, media, files
- Manage game versions and builds
- Set system requirements per game
- View developer dashboard with analytics
- Track downloads, revenue, average rating
- View game-specific reviews
- Request sales/discounts (admin approval)
- Track payout history and pending balances
- View developer-specific support tickets

ADMIN

- All permissions across platform
- Dashboard with revenue, orders, user stats
- CRUD management of all games
- CRUD management of all users (change roles, delete)
- CRUD management of categories
- View and manage all orders
- Approve/reject developer applications
- Approve/reject sale requests
- Moderate reviews (delete inappropriate)
- Manage support tickets (reply, update status, delete)

- Manage refund requests
- Manage developer profiles (demote, reactivate, delete)
- View platform reports and audit logs
- View system settings

4. User Stories

ID	Title	Description	Priority	Story Points
US-01	User Registration	As a visitor, I want to register with email and password so that I can access the platform.	High	3
US-02	User Login	As a registered user, I want to log in with my credentials so that I can access my account.	High	2
US-03	Email Verification	As a new user, I want to verify my email address so that my account is activated.	High	3
US-04	Google OAuth Login	As a user, I want to log in with my Google account so that I can skip manual registration.	Medium	5
US-05	Password Reset	As a user, I want to reset my password via email so that I can recover access if forgotten.	Medium	3
US-06	Browse Game Catalog	As a customer, I want to browse paginated game listings so that I can discover games.	High	5
US-07	Search and Filter	As a customer, I want to search by title/keyword and filter by category, price, and rating so that I can find specific games.	High	5
US-08	View Game Details	As a customer, I want to view full game details including description, screenshots, trailer, system requirements, and reviews so that I can decide whether to buy.	High	3
US-09	Add to Cart	As a customer, I want to add games to my cart so that I can purchase them later.	High	3

ID	Title	Description	Priority	Story Points
US-10	Checkout with Stripe	As a customer, I want to complete purchase via Stripe payment so that I can own the game.	High	8
US-11	Free Game Checkout	As a customer, I want to add free games to my library instantly without payment.	Medium	3
US-12	Store Credit Checkout	As a customer, I want to use store credit balance toward purchases so that I can utilize refund credits.	Medium	5
US-13	Apply Coupon	As a customer, I want to apply a coupon code at checkout so that I can get a discount.	Low	3
US-14	Write Review	As a customer, I want to rate and review games I own so that I can share my experience.	Medium	3
US-15	Manage Wishlist	As a customer, I want to add/remove games from my wishlist so that I can track titles of interest.	Low	2
US-16	View Library	As a customer, I want to view my owned games so that I can download and play them.	High	3
US-17	Download Games	As a customer, I want to download game files with version support so that I can install and play.	High	5
US-18	View Order History	As a customer, I want to view my past orders so that I can track purchases.	Low	2
US-19	Request Refund	As a customer, I want to request a refund on a completed order so that I can get my money back.	Medium	5
US-20	Friend System	As a user, I want to search users, send/accept/reject friend requests so that I can connect with others.	Medium	5
US-21	Real-Time Chat	As a user, I want to send and receive messages in real time via SignalR so that I can chat with friends.	Medium	8

ID	Title	Description	Priority	Story Points
US-22	Notifications	As a user, I want to receive and manage notifications so that I stay informed about activity.	Medium	3
US-23	Profile Posts	As a user, I want to create and delete posts on my profile so that I can share updates.	Low	2
US-24	Submit Support Ticket	As a user, I want to submit a support ticket so that I can get help with issues.	High	3
US-25	Ticket Replies	As a user, I want to reply to my support tickets so that I can follow up on issues.	Medium	3
US-26	Upload Game	As a developer, I want to upload a game with title, description, price, media, and file so that it appears in the store.	High	8
US-27	Manage Game Listing	As a developer, I want to edit or delete my game listings so that I can keep them current.	High	5
US-28	Game Versioning	As a developer, I want to upload new versions of my game so that users receive updates.	Medium	5
US-29	System Requirements	As a developer, I want to set system requirements per game so that users know minimum specs.	Low	3
US-30	Developer Dashboard	As a developer, I want a dashboard showing game count, downloads, revenue, and average rating so that I can measure performance.	High	5
US-31	Request Sale	As a developer, I want to submit a sale request with discounted price and date range so that I can run promotions.	Medium	5
US-32	View Payouts	As a developer, I want to track revenue and payouts so that I understand my earnings.	High	3
US-33	Admin Dashboard	As an admin, I want a dashboard with total users, games, orders, revenue, charts, and recent orders so that I can monitor the platform.	High	8

ID	Title	Description	Priority	Story Points
US-34	Manage Categories	As an admin, I want to create, rename, and delete categories so that the catalog stays organized.	Medium	3
US-35	Manage Users	As an admin, I want to change user roles and delete accounts so that I can enforce platform policies.	High	5
US-36	Approve Developer Applications	As an admin, I want to approve or reject developer applications so that only legitimate developers join.	High	3
US-37	Approve Sales	As an admin, I want to approve or reject developer sale requests so that discounts are controlled.	Medium	3
US-38	Moderate Reviews	As an admin, I want to delete inappropriate reviews so that platform quality is maintained.	Low	2
US-39	Manage Support Tickets	As an admin, I want to reply to, update status of, and delete support tickets so that user issues are resolved.	High	5
US-40	Manage Refunds	As an admin, I want to approve or reject refund requests so that disputes are resolved.	Medium	3
US-41	Become Developer	As a customer, I want to submit a developer application with studio name and CV so that I can become a developer.	Medium	3

5. Functional Requirements

ID	Module	Requirement	Priority
FR-01	Authentication	The system shall allow users to register with username, email, and password.	High

ID	Module	Requirement	Priority
FR-02	Authentication	The system shall allow users to log in with email and password.	High
FR-03	Authentication	The system shall send email verification upon registration and require confirmation before login.	High
FR-04	Authentication	The system shall support Google OAuth login for external account creation.	Medium
FR-05	Authentication	The system shall support password reset via email with one-hour expiry tokens.	Medium
FR-06	Authentication	The system shall support resending verification emails for unconfirmed accounts.	Medium
FR-07	Authentication	The system shall enforce rate limiting on login and registration endpoints.	High
FR-08	Profile	The system shall allow users to view their own and other users' public profiles.	Medium
FR-09	Profile	The system shall allow users to edit their avatar, bio, and email.	Medium
FR-10	Profile	The system shall allow users to change their password with current password confirmation.	Medium
FR-11	Profile	The system shall allow users to create and delete posts on profiles.	Low
FR-12	Game Catalog	The system shall display a paginated game listing on the storefront.	High
FR-13	Game Catalog	The system shall allow searching games by title and description keywords.	High
FR-14	Game Catalog	The system shall allow filtering games by category.	High

ID	Module	Requirement	Priority
FR-15	Game Catalog	The system shall show a detailed game page with description, media, system requirements, versions, and reviews.	High
FR-16	Game Catalog	The system shall display active sale prices for games on discount.	Medium
FR-17	Game Catalog	The system shall support game versioning with labels, changelogs, and file uploads.	Medium
FR-18	Cart	The system shall allow users to add games to a shopping cart.	High
FR-19	Cart	The system shall allow users to remove items from the cart.	High
FR-20	Cart	The system shall display cart item count in the navigation.	Medium
FR-21	Checkout	The system shall process payments via Stripe Checkout.	High
FR-22	Checkout	The system shall allow free checkout for games with price of zero.	High
FR-23	Checkout	The system shall support store credit as a payment method.	Medium
FR-24	Checkout	The system shall support Stripe coupons for discounts at checkout.	Low
FR-25	Checkout	The system shall handle Stripe webhook events for payment confirmation.	High
FR-26	Orders	The system shall display order history for the logged-in user.	Low
FR-27	Orders	The system shall show detailed order information including items, prices, and payment status.	Low

ID	Module	Requirement	Priority
FR-28	Library	The system shall maintain a user library of purchased games.	High
FR-29	Library	The system shall allow downloading game files with version selection.	High
FR-30	Library	The system shall support preview downloads for pre-release games when user has access.	Low
FR-31	Library	The system shall display store credit balance in the library view.	Low
FR-32	Reviews	The system shall allow users to create reviews with rating (1-5) and comment.	Medium
FR-33	Reviews	The system shall display reviews grouped by game on the storefront.	Medium
FR-34	Wishlist	The system shall allow users to toggle games in/out of their wishlist.	Low
FR-35	Wishlist	The system shall allow users to add wishlist items directly to cart.	Low
FR-36	Friends	The system shall allow users to search other users by username.	Medium
FR-37	Friends	The system shall allow sending, accepting, rejecting, and removing friend requests.	Medium
FR-38	Friends	The system shall show friend suggestions based on mutual connections.	Low
FR-39	Friends	The system shall display online status and last seen for friends.	Medium
FR-40	Chat	The system shall support real-time messaging via SignalR hub.	Medium

ID	Module	Requirement	Priority
FR-41	Chat	The system shall show unread message counts per conversation.	Medium
FR-42	Chat	The system shall mark messages as read when the conversation is opened.	Low
FR-43	Chat	The system shall show typing indicators in real time.	Low
FR-44	Notifications	The system shall create notifications for friend requests, ticket replies, and system events.	Medium
FR-45	Notifications	The system shall allow users to mark individual or all notifications as read.	Low
FR-46	Notifications	The system shall send admin-specific notifications for pending sales and developer applications.	Medium
FR-47	Support Tickets	The system shall allow users to submit support tickets with subject and message.	High
FR-48	Support Tickets	The system shall allow users and admins to reply to tickets.	High
FR-49	Support Tickets	The system shall track ticket status: Open, In Progress, Resolved, Closed.	Medium
FR-50	Support Tickets	The system shall show paginated ticket history for each user.	Medium
FR-51	Coupons	The system shall support applying Stripe coupons as discounts at checkout.	Low
FR-52	Reports	The system shall allow users to report inappropriate games or users.	Low
FR-53	Refunds	The system shall allow users to submit refund requests for completed orders.	Medium

ID	Module	Requirement	Priority
FR-54	Developer Portal	The system shall allow users to submit a developer application with studio name, description, website, and CV.	Medium
FR-55	Developer Portal	The system shall provide a developer dashboard showing game count, total downloads, total revenue, net revenue, average rating, and review count.	High
FR-56	Developer Portal	The system shall allow developers to manage game CRUD operations scoped to their own games.	High
FR-57	Developer Portal	The system shall allow developers to upload and manage game versions.	Medium
FR-58	Developer Portal	The system shall allow developers to save system requirements per game.	Low
FR-59	Developer Portal	The system shall allow developers to request sales with discounted price and date range (admin approval required).	Medium
FR-60	Developer Portal	The system shall display per-game review listings to developers.	Medium
FR-61	Admin Panel	The system shall provide an admin dashboard with total users, games, orders, revenue, free games count, charts for revenue by month, orders by day, top selling games, revenue by category, users by role, users by month, and recent orders.	High
FR-62	Admin Panel	The system shall allow admins to manage all games (create, edit, delete, add/remove screenshots, save requirements, manage versions).	High
FR-63	Admin Panel	The system shall allow admins to manage categories (create, rename, delete with constraint check).	Medium
FR-64	Admin Panel	The system shall allow admins to manage users (change role among CUSTOMER/DEVELOPER/ADMIN, delete, view details with orders/library/reviews).	High
FR-65	Admin Panel	The system shall allow admins to manage orders (view all with details).	Medium

ID	Module	Requirement	Priority
FR-66	Admin Panel	The system shall allow admins to manage reviews (view and delete).	Low
FR-67	Admin Panel	The system shall allow admins to manage support tickets (view, reply, update status, delete).	High
FR-68	Admin Panel	The system shall allow admins to manage refund requests (approve or reject with notes).	Medium
FR-69	Admin Panel	The system shall allow admins to manage developer applications (approve or reject).	High
FR-70	Admin Panel	The system shall allow admins to manage sale requests (approve or reject with reason).	Medium
FR-71	Admin Panel	The system shall allow admins to manage developer profiles (demote, reactivate, delete, edit).	Medium

6. Non-Functional Requirements

ID	Category	Requirement	Measurement Method
NFR-01	Performance	Page load time shall not exceed 2 seconds under normal load.	Lighthouse / browser DevTools timing
NFR-02	Performance	Search results shall return within 500 ms.	Server-side response time monitoring
NFR-03	Performance	The system shall support at least 1,000 concurrent users.	Load testing with k6 / JMeter
NFR-04	Security	All passwords shall be hashed with a strong algorithm (ASP.NET Core Identity default).	Code review and automated security scan
NFR-05	Security	All communications shall be encrypted via TLS 1.2 or higher.	SSL Labs test / network scan

ID	Category	Requirement	Measurement Method
NFR-06	Security	The system shall enforce rate limiting on authentication endpoints via ASP.NET Core rate limiting middleware.	API gateway metrics
NFR-07	Security	Payment data shall never be stored on application servers (Stripe handles PCI compliance).	PCI DSS compliance audit
NFR-08	Security	Anti-forgery token validation shall be enforced on all POST requests.	Automated security scan
NFR-09	Security	Admin and Developer areas shall be protected by authorization filters (AdminOnlyFilter, DeveloperOnlyFilter).	Integration tests
NFR-10	Usability	The user interface shall be consistent across all pages using shared layouts and partial views.	Design audit against style guide
NFR-11	Usability	Error messages shall be user-friendly and provide actionable guidance.	UX review and user testing
NFR-12	Usability	The system shall support responsive design for mobile and tablet viewports.	Responsive design testing (Chrome DevTools)
NFR-13	Reliability	System uptime shall be 99.9% excluding planned maintenance.	Uptime monitoring (e.g., Pingdom)
NFR-14	Reliability	Database backups shall be taken daily with point-in-time recovery.	Backup verification logs
NFR-15	Reliability	Failed transactions shall be rolled back atomically via EF Core transactions.	Integration test coverage
NFR-16	Scalability	The system shall support horizontal scaling for web tier.	Auto-scaling metrics from cloud provider
NFR-17	Scalability	Static assets shall be served via CDN or from wwwroot with caching headers.	CDN configuration audit

ID	Category	Requirement	Measurement Method
NFR-18	Scalability	Game file downloads shall support large files with streaming.	Download throughput monitoring
NFR-19	Maintainability	The codebase shall follow clean architecture with DAL, BLL, and PL layers.	Architecture review
NFR-20	Maintainability	Unit test coverage shall target at least 80% of business logic.	Code coverage reports
NFR-21	Maintainability	All external dependencies shall be managed via NuGet package references.	Dependency audit
NFR-22	Maintainability	Database schema changes shall be managed via EF Core Migrations.	Migration review

7. Requirements Traceability Matrix

FR ID	Related User Stories	Related NFRs
FR-01	US-01	NFR-04, NFR-05, NFR-08
FR-02	US-02	NFR-04, NFR-05, NFR-06, NFR-08
FR-03	US-03	NFR-04, NFR-05
FR-04	US-04	NFR-05
FR-05	US-05	NFR-04, NFR-05
FR-06	US-03	NFR-04, NFR-06
FR-07	US-01, US-02	NFR-06
FR-08	US-08	NFR-10, NFR-12
FR-09	US-09	NFR-10

FR ID	Related User Stories	Related NFRs
FR-10	US-10	NFR-04, NFR-08
FR-11	US-23	NFR-08
FR-12	US-06	NFR-01, NFR-12
FR-13	US-07	NFR-02
FR-14	US-07	NFR-02
FR-15	US-08	NFR-01
FR-16	US-06	NFR-01
FR-17	US-28	NFR-20, NFR-22
FR-18	US-09	NFR-01
FR-19	US-09	NFR-01
FR-20	US-09	NFR-01
FR-21	US-10	NFR-05, NFR-07, NFR-15
FR-22	US-11	NFR-15
FR-23	US-12	NFR-07, NFR-15
FR-24	US-13	NFR-07
FR-25	US-10	NFR-05, NFR-07
FR-26	US-18	NFR-01, NFR-10
FR-27	US-18	NFR-01
FR-28	US-16	NFR-01
FR-29	US-17	NFR-18

FR ID	Related User Stories	Related NFRs
FR-30	US-17	NFR-09
FR-31	US-16	NFR-01
FR-32	US-14	NFR-08
FR-33	US-08, US-14	NFR-01
FR-34	US-15	NFR-01
FR-35	US-15	NFR-01
FR-36	US-20	NFR-02
FR-37	US-20	NFR-08
FR-38	US-20	NFR-02
FR-39	US-20	NFR-10
FR-40	US-21	NFR-01, NFR-05
FR-41	US-21	NFR-01
FR-42	US-21	NFR-01
FR-43	US-21	NFR-01
FR-44	US-22	NFR-01
FR-45	US-22	NFR-01, NFR-08
FR-46	US-22, US-33	NFR-01
FR-47	US-24	NFR-01, NFR-08, NFR-11
FR-48	US-25, US-39	NFR-08
FR-49	US-25, US-39	NFR-10

FR ID	Related User Stories	Related NFRs
FR-50	US-24	NFR-01
FR-51	US-13	NFR-07
FR-52	US-52	NFR-08
FR-53	US-19	NFR-08
FR-54	US-41	NFR-08
FR-55	US-30	NFR-01
FR-56	US-26, US-27	NFR-08, NFR-09
FR-57	US-28	NFR-08, NFR-09
FR-58	US-29	NFR-08, NFR-09
FR-59	US-31	NFR-08, NFR-09
FR-60	US-60	NFR-09
FR-61	US-33	NFR-01, NFR-09
FR-62	US-34	NFR-08, NFR-09
FR-63	US-34	NFR-08, NFR-09
FR-64	US-35	NFR-08, NFR-09
FR-65	US-65	NFR-09
FR-66	US-38	NFR-08, NFR-09
FR-67	US-39	NFR-08, NFR-09
FR-68	US-40	NFR-08, NFR-09
FR-69	US-36	NFR-08, NFR-09

FR ID	Related User Stories	Related NFRs
FR-70	US-37	NFR-08, NFR-09
FR-71	US-71	NFR-08, NFR-09